Zijun Long

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Research Profile

Assistant Professor at Hunan University, with a PhD in Computer Science from the University of Glasgow, UK. My research develops efficient and reliable multimodal learning for information retrieval, robotic perception and industrial visual analysis. I have published **16 peer-reviewed papers as first, co-first or corresponding author**, including papers at ICML, ECCV, SIGIR and ICRA. I am lead supervisor and corresponding author of four student-led papers, have secured CNY 1,348,000 in research funding and talent support, and have independently supervised 20 MSc dissertations at Glasgow. My teaching and supervision span China and the UK.

Research interests: multimodal and interactive retrieval; reliable vision-language models; training-free and parameter-efficient adaptation; robotic perception and industrial visual learning.

Academic Appointment

Hunan University | Assistant Professor — Jan 2025 - Present

115th globally | U.S. News Best Global Universities 2026-2027

My responsibilities include research on multimodal retrieval and reliable vision-language models, teaching undergraduate and postgraduate courses, and supervising doctoral, master's and undergraduate students.

Education

University of Glasgow, UK | PhD in Computer Science — Nov 2020 - Mar 2025

77th globally | U.S. News Best Global Universities 2026-2027

Full PhD scholarship. Thesis: *Researching an Enhanced Multimodal Learning Framework for Improved Inter-Modal and Intra-Modal Alignment.* Supervisors: Richard McCreadie and Gerardo Aragon Camarasa.

University of Glasgow, UK | MSc in Data Science — Awarded Nov 2020

Entered Sep 2019; Distinction; average grade A3; ranked in the top 5% of the cohort.

South China University of Technology | BEng in Software Engineering — Awarded Jun 2019

Entered Sep 2015; Guangzhou, China; Project 985 University.

Selected Research Contributions

Multimodal retrieval. CFIR (SIGIR 2024; first author) combines entity-based filtering and summary reranking with decoupled encoding and a shared index for long-text retrieval over million-image collections. DAR (SIGIR 2025; first author) fuses refined dialogue queries with diffusion-generated visual priors, addressing text-only expression limits in multi-round text-to-image retrieval. Without task-specific fine-tuning, it improves retrieval and generalises to complex dialogue and out-of-distribution queries across four evaluated benchmarks.

Visual anomaly detection. RAD (ICML 2026; lead supervisor and corresponding author) reduces cross-category and cross-region mismatches via context-conditioned, multi-level reference retrieval. Without task-specific training, it achieved 96.7% pixel AUROC on MVTec-AD with one normal reference image per category.

Robotic foundation models and geometric perception. As first author of RoboLLM (ICRA 2024), I developed a unified multimodal framework in collaboration with Amazon, covering all three ARMBench tasks: segmentation, object identification and defect detection. As lead supervisor and corresponding author of RoboEye (ICRA 2026), I guided a method that learns when to apply 3D reranking from ranking gains and rescores candidates through keypoint correspondence confidence, using only RGB images to improve identification under occlusion and viewpoint changes.

Publications

16 peer-reviewed papers. Papers [1], [2], [3], [4]: lead supervisor and corresponding author.
* Co-first author; † Corresponding author; underlined names: students I supervise.
My name is in bold.

[1] Xingwu Zhang, Guanxuan Li, Paul Henderson, Gerardo Aragon-Camarasa, **Zijun Long**†. Is Task-Specific Training Necessary for Anomaly Detection? *ICML*, 2026. [code]

[2] Zhuocheng Zhang, Xingwu Zhang, Kangheng Liang, Guanxuan Li, Richard McCreadie, **Zijun Long**†. ADaFuSE: Adaptive Diffusion-generated Image and Text Fusion for Interactive Text-to-Image Retrieval. *SIGIR*, 2026. [code]

[3] Xingwu Zhang, Guanxuan Li, Zhuocheng Zhang, **Zijun Long**†. RoboEye: Enhancing 2D Robotic Object Identification with Selective 3D Geometric Keypoint Matching. *ICRA*, 2026. [code]

[4] Tianyi Deng, Zhaosheng Teng, Shin Takahashi, Bojun Liu, Qiu Tang, **Zijun Long**†. Regionally Routed Attention U-Net With Latent Fading Kalman Filtering for Real-Time EIT. *IEEE Transactions on Industrial Electronics*, 2026.

[5] **Zijun Long**, Kangheng Liang, Gerardo Aragon Camarasa, Richard McCreadie, Paul Henderson. Diffusion Augmented Retrieval: A Training-Free Approach to Interactive Text-to-Image Retrieval. *SIGIR*, 2025. [code]

[6] Zixuan Yi*, **Zijun Long***, Iadh Ounis, Craig Macdonald, Richard McCreadie. Enhancing Recommender Systems: Deep Modality Alignment with Large Multi-Modal Encoders. *ACM Transactions on Recommender Systems*, 2025.

[7] **Zijun Long**†, Lipeng Zhuang, George Killick, Richard McCreadie, Gerardo Aragon Camarasa, Paul Henderson. Understanding and Mitigating Human-Labelling Errors in Supervised Contrastive Learning. *ECCV*, 2024.

[8] **Zijun Long**, Xuri Ge, Richard McCreadie, Joemon M. Jose. CFIR: Fast and Effective Long-Text To Image Retrieval for Large Corpora. *SIGIR*, 2024.

[9] **Zijun Long**, George Killick, Richard McCreadie, Gerardo Aragon Camarasa. RoboLLM: Robotic Vision Tasks Grounded on Multimodal Large Language Models. *ICRA*, 2024.

[10] **Zijun Long**, George Killick, Richard McCreadie, Gerardo Aragon Camarasa. MultiWay-Adapter: Adapting Multimodal Large Language Models for Scalable Image-Text Retrieval. *ICASSP*, 2024.

[11] **Zijun Long**, Richard McCreadie, Gerardo Aragon Camarasa, Zaiqiao Meng. LaCViT: A Label-aware Contrastive Fine-tuning Framework for Vision Transformers. *ICASSP*, 2024.

[12] **Zijun Long**, Lipeng Zhuang, George Killick, Zaiqiao Meng, Richard McCreadie, Gerardo Aragon-Camarasa. CLCE: An Approach to Refining Cross-Entropy and Contrastive Learning for Optimized Learning Fusion. *ECAI*, 2024.

[13] **Zijun Long**, Richard McCreadie, Muhammad Imran. CrisisViT: A Robust Vision Transformer for Crisis Image Classification. *ISCRAM*, 2023.

[14] **Zijun Long**, Richard McCreadie. Is Multi-Modal Data Key for Crisis Content Categorization on Social Media? *ISCRAM*, 2022.

[15] **Zijun Long**, Richard McCreadie. Automated Crisis Content Categorization for COVID-19 Tweet Streams. *ISCRAM*, 2021.

[16] **Zijun Long**, Xiaohang Wang, Yingtao Jiang, Guofeng Cui, Li Zhang, Terrence Mak. Improving the Efficiency of Thermal Covert Channels in Multi-/Many-core Systems. *DATE*, 2018.

Research Funding, Awards and Projects

Hunan Provincial Department of Science and Technology | Award Recipient — Awarded 2026

Hunan Furong Program Young Talent Award (Science and Technology Innovation).

CNY 500,000 | Funding period: Jul 2026 - Jul 2029

State Grid Corporation of China | Principal Investigator — Mar 2025 - Sep 2026

CNY 848,000 | Completed

Reliability Testing Technology and AI Models for Operational Data Analysis of Electricity Metering Equipment under Multiple Extreme Environmental Conditions.

University of Glasgow, UK | Research Contributor

Chemobots: Digital Chemical Robotics (EPSRC EP/S019472/1): contributed to the optimisation of robotic control algorithms supporting the automation of molecular-synthesis workflows.

BigDataStack (EU Horizon 2020, No. 779747): contributed protocols for dynamic allocation of resources in cloud-edge computing. **INFINITECH** (EU Horizon 2020, No. 856632): contributed algorithms for autonomous service orchestration and scalable testbed integration in real-time financial analytics. I also contributed research software and experiments to EPSRC-funded projects on robotic grasping.

Student Supervision and Research Leadership

Student supervision at Hunan University: **one PhD student, one master's student and two undergraduates**, including Yixiao Liu (PhD) and Junshu An (Master's). Broader research mentoring: five graduate and eight undergraduate students. Lead supervisor and corresponding author of four student-led papers published in 2026; the supervised students and their papers are identified below.

ICML 2026, RAD [1]: Xingwu Zhang and Guanxuan Li.
SIGIR 2026, ADaFuSE [2]: Zhuocheng Zhang, Xingwu Zhang, Kangheng Liang and Guanxuan Li.
ICRA 2026, RoboEye [3]: Xingwu Zhang, Guanxuan Li and Zhuocheng Zhang.
IEEE TIE 2026, real-time EIT [4]: Tianyi Deng.

University of Glasgow | MSc Project Supervisor — May 2023 - Oct 2023

Independently supervised 20 AI-focused MSc dissertations, each accounting for one-third of the degree credits. Six received A grades and ranked within the top 20% of the overall assessed cohort.

Teaching Experience

Hunan University

Teach three courses to 200+ students annually, with 32 contact hours per course.

Intelligent Information Processing — Mar-Jun 2026
Graduate

Professional Writing and Scientific Communication — Since Sep 2026
Undergraduate

Satellite Internet Applications for New-Type Power Systems — Since Sep 2026
Undergraduate

University of Glasgow | Teaching Assistant (Tutor and Marker) — Nov 2020 - Jun 2024

Tutored and assessed coursework and exams in Recommender Systems, Text as Data, Introduction to Data Science, Programming, and Machine Learning, providing tutorials and constructive feedback.

Awards and Honours

Furong Program Young Talent 2026 (funding above)
(Science and Technology Innovation), Hunan Province

Selected for a provincial talent award recognising research achievement and innovation potential.

SPS Travel Grant | IEEE Signal Processing Society | £500 2024

RSC Travel Grants | University of Glasgow 2022 - 2024
£15,250 across three awards

Best Student Paper Award | ISCRAM, Blacksburg, Virginia, USA 2021

Full PhD Scholarship | University of Glasgow | £150,000 2020

First Class Scholarship | South China University of Technology 2018

Model Training and Research Computing

LLM post-training for retrieval. Implemented an end-to-end Llama 3 post-training pipeline for query reformulation and candidate reranking, using supervised fine-tuning (SFT) and proximal policy optimization (PPO). Tested a reward model trained on human preferences and rewards based on retrieval metrics.

Distributed training and cluster operations. Ran distributed training across 120 GPUs and performed cluster operations including job submission and queue management, resource allocation, software environment deployment, GPU troubleshooting, checkpoint recovery and experiment monitoring.

Professional Service

Proceedings Chair | British Machine Vision Conference (BMVC) 2024

Program Committee | AAAI (since 2023); ISCRAM (since 2022)

Conference Reviewer 2022 - Present
CVPR, ECCV, SIGIR, ICML, NeurIPS, EMNLP, ACL

Journal Reviewer | ACM Transactions on Information Systems; Knowledge-Based Systems

Professional Memberships | IEEE; ACM